Sanya Gupta

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EDUCATION

University of Engineering & Technology, Panjab University — Chandigarh
Bachelors of Engineering in Information Technology | GPA 8.18 — *July 2023- August 2027*

GGMSSS 20B — Chandigarh, India
PCM CBSE BOARD | ***93%*** — *April 2021- April 2023*

TECHNICAL SKILLS

Generative AI & LLMs: LLMs (Transformers, Hugging Face), RAG (LangChain, FAISS), Computer Vision (OpenCV,YOLO), TensorFlow, PyTorch, scikit-learn, NumPy.
Backend & APIs: Python, FastAPI, Flask, RESTful APIs, JWT/OAuth 2.0, Asyncio
Database: PostgreSQL (JSONB/Vector storage), MySQL.,pgvector
Frontend & Design: JavaScript, React, Tailwind CSS, Data Visualization.
Tools & DevOps: Docker, CI/CD, Linux, Git, Google Cloud Platform (GCP), Jupyter, MATLAB, C++.

EXPERIENCE

AISoC [AI Summer of Code] — June 2025 - July 2025
Open Source Contributor | Python, Deep Learning, YOLO

- Deployed a real-time vehicle detection system using **YOLO**
- Optimized YOLO-based inference for low-light environments, improving precision to 0.82 through hyperparameter tuning

Design Innovation Center, PU — May. 2024 – Jan. 2025
Deep Learning Intern | Python,DL architectures

- Spearheaded experimentation **with 3D Architectures (U-Net, V-Net, and GANs)** on a novel skull dataset, optimizing the training loop to significantly reduce computation time
- Developed an automated Python pipeline leveraging Connected Component Analysis (CCA) to refine segmentation masks, entirely eliminating noise and significantly boosting model precision.

PUBLICATION

[SCAI-NET] - ***An AI-Driven Framework for Optimized, Fast, and Resource-Efficient Skull Implant Generation for Cranioplasty using CT Images***, *published in Journal of Computers in Biology and Medicine [CS-13, **Impact Factor: 7**]*

PROJECTS

Cite Craft| [website]| *React, FastAPI, PostgreSQL, RAG, Transformers*

- Designed a section-aware RAG tool to improve researcher productivity. Prioritized feature development for 'Source Attribution' based on user need for 100% verifiability in AI-generated reports
- **Developed a multi-source aggregator** integrating **5 academic APIs** (including Semantic Scholar and arXiv); engineered a data pipeline for automated metadata extraction and **CSV-based data portability** for large-scale literature reviews.
- Engineered a **comparative analysis engine** featuring per-cell RAG retrieval; implemented **direct source attribution** to display original paper snippets, ensuring 100% verifiability of AI-generated comparisons.
- Orchestrated a high-performance backend with **FastAPI** and **PostgreSQL (JSONB)** for efficient embedding storage; implemented **JWT-based secure access** and managed a multi-cloud deployment via Railway and Vercel.
- Optimized for high-performance retrieval, achieving **sub-500ms response times** for RAG-based queries through efficient **PostgreSQL JSONB indexing** and asynchronous FastAPI endpoints.

AutoOutreach | [code] | *Python, Google OAuth, Hugging Face*

- Developed an **automated lead generation and outreach engine** using Python, integrating **Google OAuth 2.0** and **Google Sheets API** for dynamic data synchronization.
- Leveraged **Hugging Face Inference APIs** to perform zero-shot analysis of job descriptions, generating highly personalized email bodies via tailored prompt templates.

Automated Deployment Pipeline | [website] |Docker, GitHub Actions

- Developed a **framework-agnostic deployment script** that programmatically detects project frameworks (Python, JS, etc.) to auto-generate optimized **Dockerfiles** and **.dockerignore** files.
- Integrated with GitHub Actions to provide a zero-touch CI/CD experience, reducing manual configuration time for new microservices to near zero.